

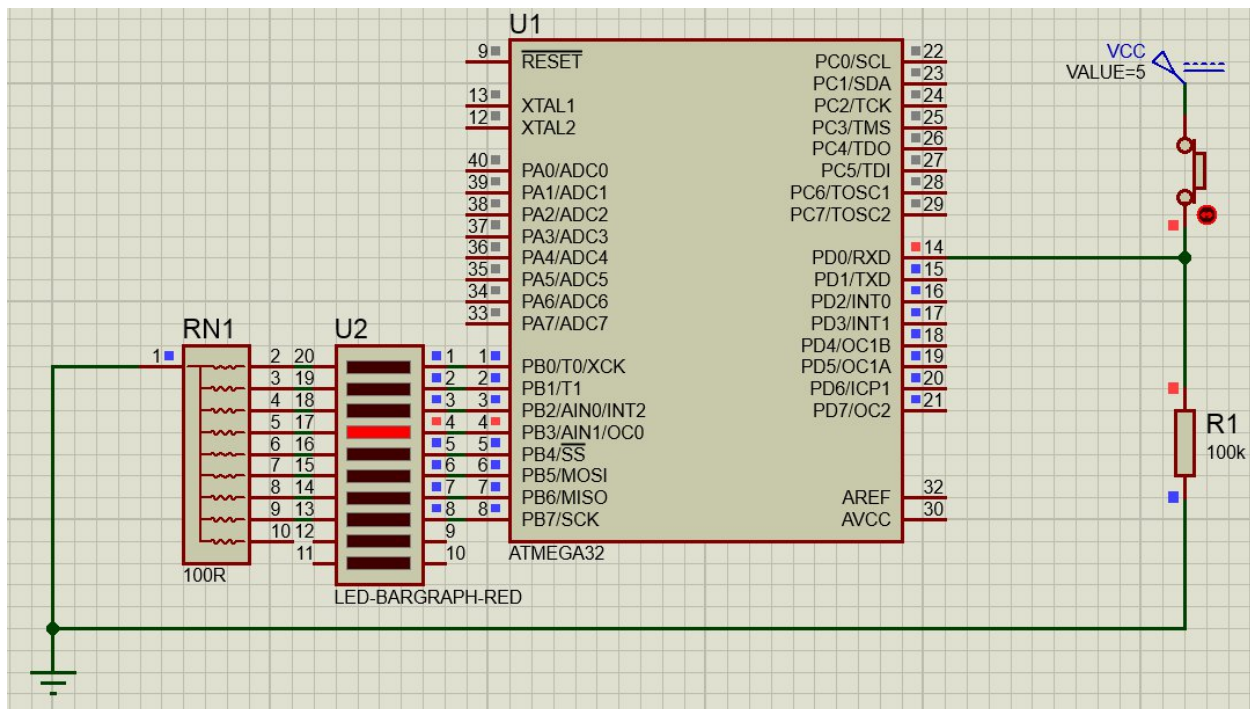
ME 2261 - Embedded Systems Log Book

Using break function with button

```

1  #define F_CPU 8000000UL // 8 MHz clock frequency
2  #include <avr/io.h>
3  #include <util/delay.h>
4
5  int main(void)
6  {
7      DDRB = 0xFF;
8      DDRD = 0xFE;
9      PORTD = 0;
10     for (int a=1;a<130;a=a*2)
11     {
12         if (PIND==1){
13             _delay_ms(500);
14             break;
15         }
16         PORTB = a;
17         _delay_ms(1000);
18     }
19     return 1;
20 }

```



Using 'continue' statement

```
1 | #include <avr/io.h>
2 |
3 | int main(void)
4 | {
5 |     int i;
6 |     for (i=0;i<10;i++)
7 |     {
8 |         if (i%2==0){
9 |             continue;// Skip the even values of i
10 |        }
11 |        PORTA = i;
12 |    }
13 |    return 1;
14 | }
```

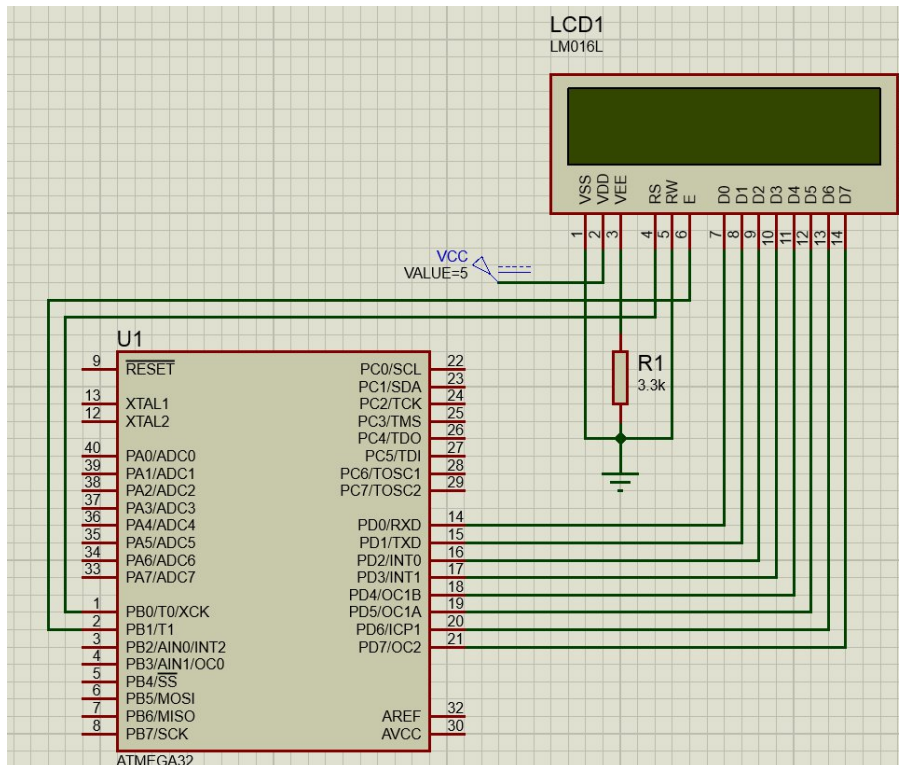
Using Macros

```
1 | #include <avr/io.h>
2 |
3 | int main()
4 | {
5 |     DDRD = 0xFF;
6 |     PORTD = 0;
7 |     PORTD = _BV(0)|_BV(2)|_BV(7);
8 |     PORTD = 0xFF;
9 |     PORTD &= ~(_BV(1)|_BV(2)|_BV(6));
10 |    PORTD = 0;
11 |    PORTD = _BV(2)|_BV(4)|_BV(6);
12 |    while(1){
13 |        PORTD ^=(_BV(2)|_BV(4)|_BV(6));
14 |    }
15 |    return 1;
16 | }
```

Task 1

LCD Display

```
1  #define F_CPU 8000000UL // 8 MHz clock speed
2  #include <avr/io.h>
3  #include <util/delay.h>
4  #define D0 eS_PORTD0
5  #define D1 eS_PORTD1
6  #define D2 eS_PORTD2
7  #define D3 eS_PORTD3
8  #define D4 eS_PORTD4
9  #define D5 eS_PORTD5
10 #define D6 eS_PORTD6
11 #define D7 eS_PORTD7
12 #define RS eS_PORTB0
13 #define EN eS_PORTB1
14 #include "lcd.h"
15
16 int main(void){
17     DDRD = 0xFF;
18     DDRB = 0xFF;
19     Lcd8_Init();
20     while(1){
21         Lcd8_Set_Cursor(1,0); // led.setCursor(col,row)
22         Lcd8_Write_String("NEW Embedded LAB"); // Maximum 16 Characters can be added
23         Lcd8_Set_Cursor(2,1);
24         Lcd8_Write_String("*** 2025 ***");
25     }
26 }
```

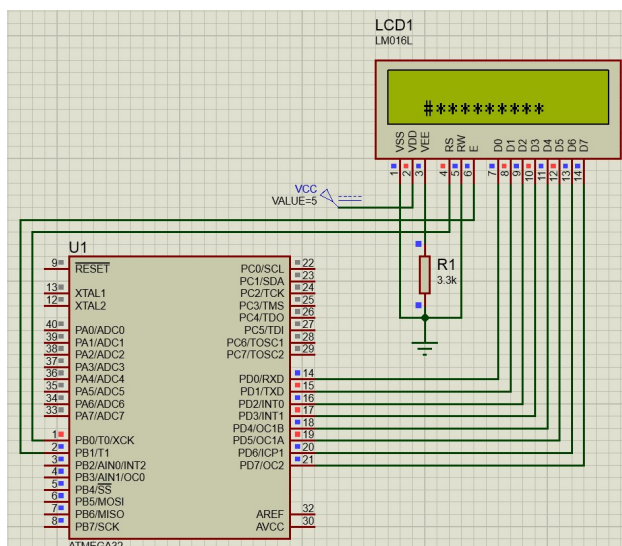


Task 2

```

16 int main(void){
17     DDRD = 0xFF;
18     DDRB = 0xFF;
19     Lcd8_Init();
20     while(1){
21         Lcd8_Set_Cursor(1,2);
22         Lcd8_Write_String("Embedded LAB");
23         Lcd8_Set_Cursor(2,3);
24         Lcd8_Write_String("*** 2024 ***");
25         _delay_ms(1000);
26         Lcd8_Clear();
27
28         Lcd8_Set_Cursor(2,0);
29         Lcd8_Shift_Right();
30         Lcd8_Shift_Right();
31         Lcd8_Write_Char('#');
32         _delay_ms(200);
33         for (int i=0;i<10;i++)
34         {
35             Lcd8_Write_Char('*');
36             _delay_ms(200);
37         }
38         Lcd8_Write_Char('#');
39         _delay_ms(500);
40         Lcd8_Clear();
41
42         Lcd8_Set_Cursor(1,0); // led.setCursor(row[1,2],col[0,15])
43         Lcd8_Write_String("NEW Embedded LAB"); // Maximum 16 Characters can be added
44         Lcd8_Set_Cursor(2,3);
45         Lcd8_Write_String("*** 2025 ***");
46         _delay_ms(1000);
47         Lcd8_Clear();
48         _delay_ms(500);
49     }
50 }

```

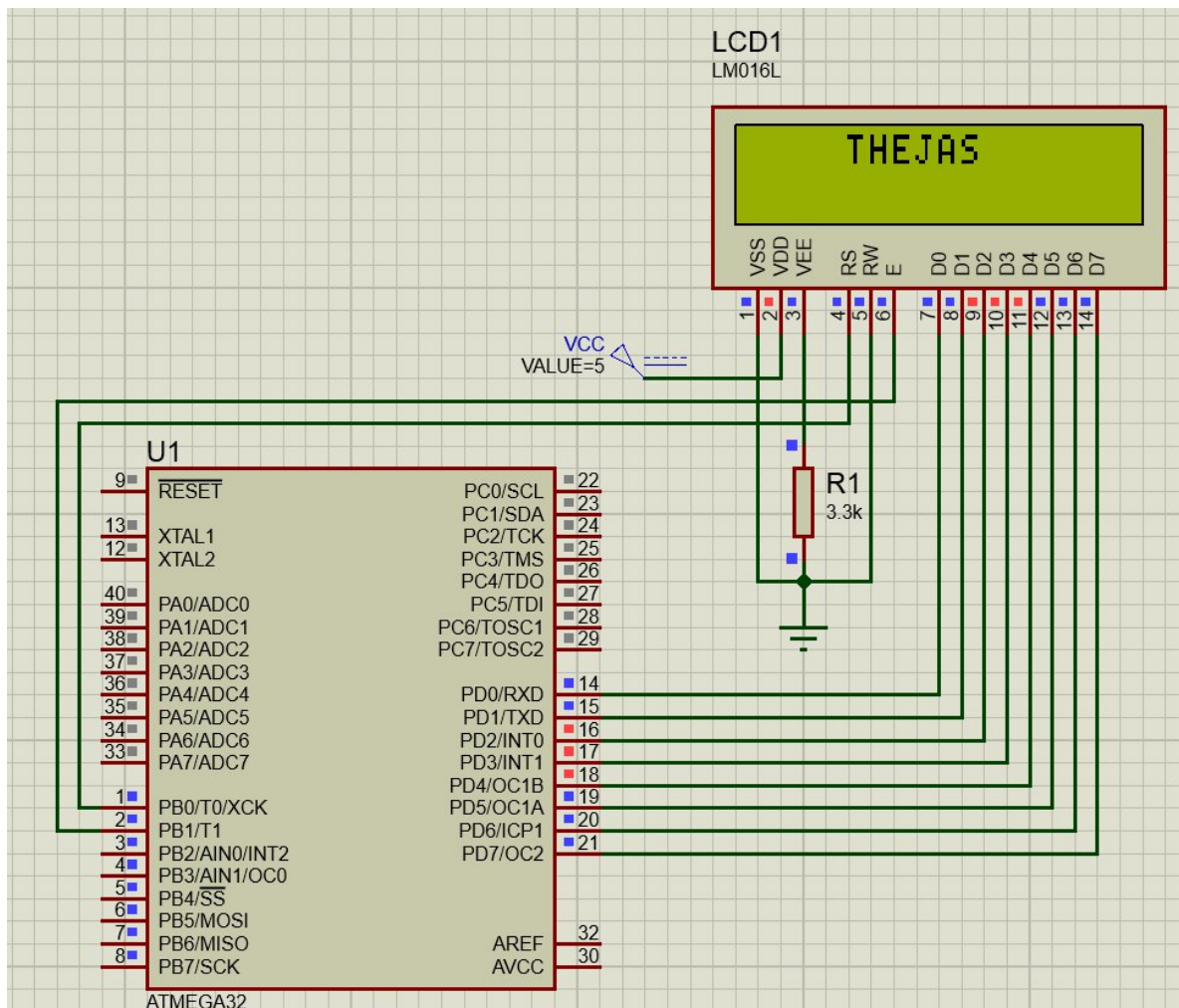


Task 3

```

16 int main(void){
17     DDRD = 0xFF;
18     DDRB = 0xFF;
19     Lcd8_Init();
20     Lcd8_Set_Cursor(1,0);
21     Lcd8_Write_String("THEJAS");
22     while(1){
23         for (int i=0;i<10;i++)
24         {
25             Lcd8_Shift_Right();
26             _delay_ms(700);
27         }
28         for (int i=0;i<10;i++)
29         {
30             Lcd8_Shift_Left();
31             _delay_ms(700);
32         }
33     }
34 }

```



Task 4

String goes left until finish the sentence.

```
16 int main(void){
17     DDRD = 0xFF;
18     DDRB = 0xFF;
19     Lcd8_Init();
20     while(1){
21         Lcd8_Set_Cursor(1,0);
22         Lcd8_Write_String("NEW Embedded LAB 2025");
23         for (int i=0;i<5;i++)
24         {
25             _delay_ms(1000);
26             Lcd8_Shift_Left();
27         }
28         _delay_ms(1000);
29         Lcd8_Clear();
30         _delay_ms(100);
31     }
32 }
```

Task 5

String goes left until all sentence vanish.

```
16 int main(void){
17     DDRD = 0xFF;
18     DDRB = 0xFF;
19     Lcd8_Init();
20     while(1){
21         Lcd8_Set_Cursor(1,0);
22         Lcd8_Write_String("NEW Embedded LAB 2025");
23         for (int i=0;i<21;i++)
24         {
25             _delay_ms(1000);
26             Lcd8_Shift_Left();
27         }
28         _delay_ms(1000);
29         Lcd8_Clear();
30         _delay_ms(100);
31     }
32 }
```